QI LING

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3rd-year Ph.D. student in Computer Science, Purdue University

RESEARCH INTERESTS

I want to make emerging workloads and secure systems more efficient, exploring co-design across architecture, compilers, runtime systems, and applications. Currently, my work focuses on efficient serverless computing, agentic workloads, and mitigations for attacks.

EDUCATION

- **Purdue University** **Aug. 2024 – Present**
  Doctoral Degree in CS **Overall GPA: 4.00/4**
  Advisors: Prof. Kazem Taram and Prof. Pedro Fonseca
- **University of Michigan** **Sept. 2022 – May 2024**
  Bachelor's Degree in CS **Overall GPA: 3.92/4**
  Advisors: Prof. Baris Kasikci and Prof. Shuwen Deng
- **Shanghai Jiao Tong University** **Sept. 2020 – Aug. 2024**
  Bachelor's Degree in ECE **Overall GPA: 3.74/4**
  Advisor: Prof. Yibo Pi

HONORS AND SERVICES

- **USENIX Security Symposium Program Committee** **2027**
- **USENIX Security Distinguished Paper Award** **2026**
- **ACM MICRO Student Research Competition 2nd Place Award** **2023**
- **Teaching Assistant for Computer Architecture, Security, and Algorithm** **2024 - 2026**

PUBLICATIONS AND MANUSCRIPTS

- **Characterizing and Improving Agentic Workloads on CPUs** **In progress**
- **Warming Up Cold Branch Predictors in Serverless** **In submission**
- **A Harmony of Ice and Fire: Mitigating Cold Last-Level Cache Effects in Serverless Computing on Commodity Hardware**
  Qi Ling, Ajay R. Rawat, Pedro Fonseca, Kazem Taram
  59th IEEE/ACM International Symposium on Microarchitecture (**MICRO'26**)
- **An Integer Overflow Endgame: Compiler and Architecture Support for Default-On Integer Overflow Mitigation**
  Zheng Zhang, **Qi Ling**, Kian Kasad, Brendan Ryan Sweeney, Deepak Gupta, Tal Garfinkel, Kazem Taram
  35th USENIX Security Symposium (**USENIX Security'26**)
  Distinguished Paper Award
- **GadgetMeter: Quantitatively and Accurately Gauging the Exploitability of Speculative Gadgets**
  Qi Ling, Yujun Liang, Yi Ren, Baris Kasikci, Shuwen Deng
  Network and Distributed System Security Symposium (**NDSS'25**)
  2nd Place at MICRO Student Research Competition

RESEARCH EXPERIENCE

Making Emerging Workloads Efficient on Current Hardware

- **Characterizing and improving agentic AI workloads on CPUs** **2026 – Present**
  - Characterizing where end-to-end time goes in LLM agents beyond model inference, including tool execution, sandboxing, and orchestration on CPUs.
  - Identifying how agentic execution differs from conventional workloads, to guide CPU optimizations.
- **Eliminating CPU cold-start penalty in serverless computing** **2024 – Present**
  - Identified the last-level cache (LLC) and conditional branch predictor (CBP) as the structures behind over 95% of serverless functions' CPU cold-start penalty.
  - Proposed novel techniques to partition the LLC and CBP across serverless functions, reducing overhead by 7× on LLC and 9× on CBP.
  - Designed scheduling algorithms that allocate LLC and CBP to functions to maximize overall performance, speeding up latency-critical functions by up to 65% (34% on average).

Making Secure Systems Efficient

- **Enabling low-cost default-on integer overflow mitigation** **2024 – 2026**
  - Identified two sources of overhead: checking every source-level arithmetic operation blocks compiler optimizations, though many operations can be optimized and need no check; further, current ISAs double the µops of checked arithmetic.
  - Proposed a new compiler technique that defers overflow checks until after compiler optimizations, eliminating unnecessary checks and reducing overhead on SPEC CPU 2017 from 13.9% to 3.6%.
  - Proposed ISA extensions that issue overflow-checked arithmetic as a single µop, reducing overhead further from 3.6% to 0.7%.
- **Enabling precise, low-cost software Spectre mitigation** **2023 – 2025**
  - Identified timing condition, a race between branch resolution and secret leakage, as a fundamental requirement for an exploitable Spectre gadget.
  - Proposed a gadget detection technique that models a gadget's timing condition as an instruction DAG, simulates the strongest attacker, and quantifies exploitability through runtime testing.
  - Reduced the false-positive rate of gadget scanners from 51.1% to 0.6% across 6 applications and the Linux kernel, cutting patching overhead by up to 44.1% (20.7% on average).

TECHNICAL SKILLS

- **CPU design**: Verilog, Gem5
  - Implemented a P6 OoO processor with speculative execution and prefetchers.
- **CPU microarchitectural analysis**
  - Identified program bottlenecks in CPU microarchitecture.
  - Characterized Spectre vulnerabilities and microarchitectural side channels.
- **LLVM compiler development**
  - Added frontend and backend support for new overflow instructions.
  - Developed middle-end passes to accurately identify and patch Spectre vulnerabilities.
- **Linux kernel development**
  - Implemented page coloring (1k LoC) within Linux memory management subsystem.
  - Implemented an ultra-fast NAT router with an eBPF module (GitHub).
- **Others**: Honggfuzz, Syzkaller, C/C++, Python, assembly, Kubernetes